TECHNICAL REPORT FOR CADENZA ICASSP 2024 GRAND CHALLENGE

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ABSTRACT

This report briefly demonstrates our approach on *Cadenza ICASSP 2024 Grand Challenge*. For the music source separation task, deep learning models trained on the MusDB18 dataset such as Demucs or OpenUnmix achieved impressive results on MusDB18. However, these methods use additive signals as input to the model to obtain anechoic multi-track signals directly. For the reverberation environment, we use personalized reverberation data to fine-tune the separation model to obtain the output of each audio track.

***Index Terms*—** Pretrained model, Head related transfer function, Music Source Separation, Finetune

1. INTRODUCTION

Cadenza Grand Challenge, part of *ICASSP 2024 Signal Processing Grand Challenge 2024*, is a challenge aiming at developing a signal processing system that allows personalised rebalancing of the music to improve the listening experience, e.g. amplifying the vocals with respect to the sound of the band. One approach would be to separate each track and then apply gains to the separated tracks to rebalance when the music is remixed as a stereo signal. Compared with conventional music separation task, this task is closer to the real-world circumstance, i.e. the stereo signals are replayed via the loudspeakers, and the respective head-related transfer functions (HRTF) are convoluted with the original signals to obtain the signals at both ears.

In Section 2, We propose our approach of fine-tuning of the pre-trained Hybrid Demucs [1, 2] model oriented to the given acoustic environments. The overall architecture of the network adopts the encoder-decoder paradigm, and the outputs are fixed to four tracks: vocal, bass, drum, and other. We demonstrate our experimental setup and the results of our proposed methods in Section 3.

2. METHODS

The overall architecture of our proposed method is demonstrated in Fig. 1. In the original Hybrid Demucs model, each source is represented by a waveform $s_i \in R^{C,T}$ where C is the number of channels (where 2 is set for stereo) and T is the number of samples. For a waveform x_s containing T samples and corresponding to source s, a predicted waveform $\hat{x}_s$ and denoting with a subscript t the t-th sample of a waveform, $L1$ loss is chosen to supervise in the wave-domain as

$$L_1(\hat{x}_s, x_s) = \frac{1}{T}\sum_{i=1}^{T} |\hat{x}_{s,t} - x_{s,t}| \tag{1}$$

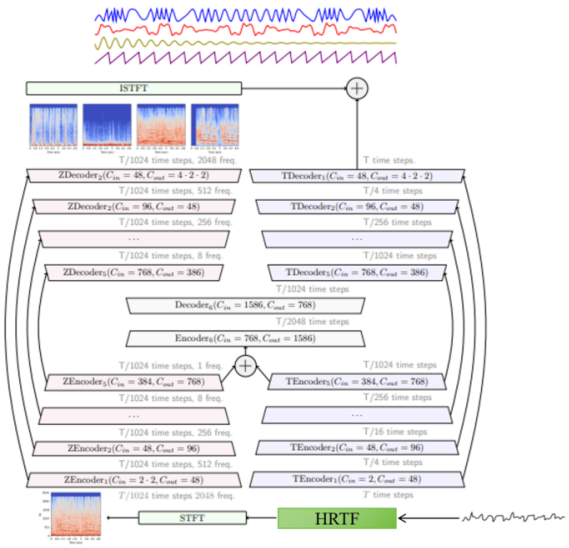

Fig. 1. Music Sound Source Separation With Demucs.

The original Hybrid Demucs model is trained with clean audio waves with high fidelity. Nevertheless, under certain acoustic conditions, the stereo signal at the receiving end is somewhat different and more complex due to the combined presence of direct sound and the reverberated sound, which may cripple the performance of the original model. To deal with this problem, We propose two systems, demonstrated as below.

Method 1 We fine-tune the Hybrid Demucs model with all training data after applying HRTFs to them, without distinguishing the directions of the HRTFs, and We take advantage of the Hybrid Demucs model by applying the given convo-

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luted HRTF stereo mixture data as input and the given convoluted HRTF four stereo tracks as output. The whole fine-tuning process takes about 20 epochs in total, which means that we think of HRTF as a mean of enhancing the data and let Hybrid Demucs itself fit the new data in this case. We do not distinguish the combination direction of the HRTF, and fine-tune all training data after applying the transfer function.

Method 2 When the two loudspeakers are placed as $(-22.5^o, +22.5^o)$, $(-30^o, +30^o)$ or $(-37.5^o, +37.5^o)$, the listener will perceive the sound as coming from the center all the same, and thus may be regarded as a same case. Based on this fact, we propose a direction-oriented fine-tuning approach. We simplify the combination of different loudspeaker orientations to 4 individual cases and implement the system based on this prerequisite. For example, when the orientation combination is $(-30^o, +22.5^o)$, we can exchange the two channels and use the same network training as $(-22.5^o, +30^o)$. This may also help reducing overfitting for the symmetrical cases. The configurations of the loudspeaker direction are shown in Table 3.

3. EXPERIMENTS & RESULTS

As recommended by the authors of Hybrid Demucs, we use Dora for model fine-tuning[1]. We choose the *81de367c* pre-trained model as the base model for fine-tuning, which is trained based on MUSDB18-HQ[2]. It contains 48ch hybrid model with residual branches, group norm, singular value penalty and amplitude spectogram. We use the mixed binaural signals as input and the four-track stereo signals as output, with HRTFs applied according to the officially given settings.

According to the given metadata, there are sound sources from 3 directions on the left and right sides, which can yield a set of 9 pairs of sound source combination, as illustrated in Fig. 2.

Table 1. Finetune Datatype

ID	Left	Right
1	22.5	30
2	22.5	37.5
3	30	37.5
4	Symmetry	

We used the official training set to generate a total of 400-song data corresponding to different head and loudspeaker positions and divided the training and validation sets by the ratio of 8:2 for experiments. During the training process, we did not use other datasets or augment the original data, nor did we use the official validation set or test set as training data. Method 1 takes 20 epochs to converge, while method 2 takes 20-40 epochs for each individual models. We utilized 4 GeForce RTX 4090 GPUs for training are chosen for distributed learning and batch size is set as 16. We test our model on the validation set, and enhanced the audios from test set based on the sound source direction corresponding to the test set. The expeimental results are shown in Table 3.

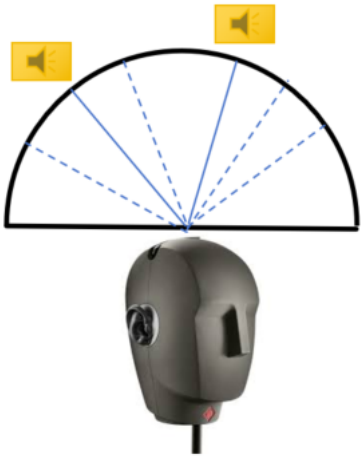

Fig. 2. Experiments Condition.

Table 2. Results on the valid set.

Method	HAAQI
Baseline	0.668
Method 1	0.615
Method 2	0.604

The enhanced signals of method 1 are packed in submission_T042.zip and the enhanced signals of method 2 are packed in submission_T042_aug.zip

4. REFERENCES

[1] Alexandre Défossez, Nicolas Usunier, Léon Bottou, and Francis Bach, "Music source separation in the waveform domain," *arXiv preprint arXiv:1911.13254*, 2019.

[2] Simon Rouard, Francisco Massa, and Alexandre Défossez, "Hybrid transformers for music source separation," in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.

[1] https://github.com/facebookresearch/dora

[2] https://github.com/facebookresearch/demucs/blob/main/docs/training.md